Research Statement

Alireza Azampour
Institute for International Economic Studies, Stockholm University
alireza.azampour@iies.su.se

I study firm decisions in quantitative macroeconomic models when energy, debt, and technology are not a smooth margin, and when a linear impulse response would miss the sign or the size of the shock. The projects below share that question. The job market paper and the work on forward guidance also require a nonlinear transition. There I check the linear approximation by comparing impulse responses frequency by frequency, in the z-domain.

Job market paper

Oil Price Volatility and the Asymmetric Transmission of Energy Shocks, with Mohammad Naderitabar.

Adverse oil-price movements contract activity by more than favorable movements expand it, and the gap grows with the size of the shock. Part of the shock is the new price level. Part of it is the volatility that comes with the move. We study that second part in a heterogeneous-agent model with two sectors, one more oil-intensive than the other. Volatility reaches aggregates through firm cash flows, through the risk of a single firm's equity, and through the concentrated portfolios households choose.

Firms commit before the oil price is known. They choose capital and planned variable inputs, and they finance operating inputs, before revenue arrives. An idiosyncratic revenue shock is drawn after those commitments. A low enough draw liquidates the firm's capital. Liquidation hits one firm, so its return is more volatile than a diversified claim on the whole sector.

Households hold the concentrated claim. They remain familiar with the sector in which they work, so a firm in that sector is the natural investment, and spreading risk across firms and sectors is costly. The same household therefore receives labor income from a sector and bears that sector's equity risk. When oil-price volatility rises, cash-flow dispersion and downside risk widen, especially in the oil-intensive sector. Required returns rise, and investment falls in both sectors, by more where oil intensity is high. Decreasing returns in variable inputs, costly adjustment of those inputs, and liquidation losses that depend on the state then turn the smaller capital stock into lower productive capacity and lower wages.

An increase and a decline are different events, because volatility rises either way. After an increase, higher costs, weaker cash flow, and greater liquidation risk arrive while households are less willing to bear risk. After a decline, lower marginal costs would expand activity, but higher return volatility still discourages concentrated investment, so the expansion is damped. Dispersion, liquidation, and investment sensitivity should move most among oil-intensive firms. A representative firm facing only a larger oil bill, or a real option that ignores sectoral equity risk, does not produce that ranking.

National accounts discipline aggregate labor compensation and oil expenditure. The revenue-shock support and the liquidation loss are set to a quarterly liquidation probability of 1.5 percent and a capital-weighted idiosyncratic return variance of 0.0038. MIT experiments move the price level, its sign, and conditional volatility, separately and jointly. Each path is solved with a standard nonlinear method and with an almost-nonlinear level-k algorithm. Given beliefs about the aggregate path, household and firm policies are recomputed nonlinearly, and on-impact derivatives correct only small within-period price errors.

Debt portfolios and firm life cycle

The job market paper follows the risk of owning a firm through to investment and wages. The next paper starts from the debts the firm itself issues, and follows them through to employment.

Firm Debt Portfolios and Factor Adjustment, with Mikael Carlsson, Alireza Sepahsalari, and Oskar Nordström Skans.

Swedish firm-level data show a split in the use of debt. Risky short-term debt finances working capital. Safe long-term debt finances growth. The model gives the two instruments different jobs. Productivity, long-term debt, and capital are the state variables. Long-term debt is collateralized by capital, so it is what accompanies a larger capital stock. Short-term debt pays for inputs inside the period and is due after a production shock. If revenue and a partial sale of capital cannot cover that payment, the firm defaults. Capital is costly to adjust outside an inaction band, and the long-term debt stock is costly to change, so neither position is reset every period.

We match how capital, employment, sales, and debt structure evolve with firm age, and how those variables line up in the cross-section. We then change the supply of short-term debt relative to long-term debt, in the steady state and along the transition after an aggregate shock, and follow the response of hiring.

Technology adoption in the power sector

The first two papers concern financial positions that are costly to adjust. In electricity generation the position is technological: the fuel a plant burns, its efficiency, and the years since that efficiency last changed. What the generator data show is separate from what the equilibrium does with those facts.

Technology Adoption and the Pace of Transition in the U.S. Power Sector, with Maija Kaartinen.

Empirical work. From the early 2000s through the 2010s, U.S. generation moved from coal toward gas and then toward wind and solar. The EIA generator panel for 1991–2021 measures, at the unit, how plants that burn a fuel adopt. Four facts describe that decision. Between updates, thermal efficiency falls with the years since the last improvement. Efficiency persists from year to year and across an update. The probability of an update rises with time since the previous one, and falls when that update left the plant efficient. An update is an efficiency gain of at least 10 percent of which at least 40 percent remains the next year. The facts hold in the full sample and among single-generator plants. Wind and solar have no heat rate of this form. They appear in the evidence as aggregate generation and as a declining installed cost.

Model. The plant's state is built to carry these facts. It is a coal, gas, or intermittent-renewable technology, together with an efficiency draw and a technological age. Age lowers rated efficiency when there is no update. A new draw is correlated with the current draw. Adoption is a discrete payment, so a plant waits while its draw is still good and pays once the draw has depreciated. Within the year the plant chooses hourly utilization. For a thermal unit, efficiency is low at low load and reaches the rated level only near full capacity. Across years the plant may adopt and reset part of its age, or exit. Entrants pay a technology-specific cost. Coal and gas buy different fuels, each along its own upward-sloping supply. A renewable plant buys no fuel. Output is capacity times availability, and renewable availability is lower when the hourly electricity price is high.

Utilization against efficiency is estimated from the generator data. Construction costs, fixed costs, and the efficiency of new units come from the EIA, and long-run demand and fuel-supply elasticities from the literature. The remaining scales match the initial coal-and-gas steady state. A perfect-foresight path then brings in cheaper combined-cycle capital, more efficient new gas units, the shale increase in gas supply, and later cheaper renewable capital. The fossil portion checks

whether entry, exit, and adoption can move the generation mix about as much as in the data.

On the renewable portion, the policy comparison is an experiment in the equilibrium. A carbon tax raises the fuel cost of plants already operating, and so changes utilization, exit, and adoption. An installation subsidy lowers the cost of renewable entry and of a new efficiency draw. Each instrument is run at 5 percent and at 20 percent of benchmark scale, for one year, and at the smaller scale for ten years. Inherited efficiency and age make the one-year path differ from the ten-year path. For each path we record the renewable share of generation, the fiscal transfer, the change in electricity expenditure, and the net cost to consumers.

Forward guidance in HANK models

Forward guidance can lead to a nonlinear transition path. The shock is news, so decisions are made before the policy arrives.

Revising Forward Guidance and Targeted Policies in HANK Models Using an Almost Nonlinear Solution, with Mohammad Naderitabar.

The setting is a heterogeneous-agent New Keynesian model, with sticky prices. Forward guidance and targeted transfers are announcements, so their effect depends on policy functions away from the steady state and on the distribution. We recompute both with the level-k method of the job market paper and set that path next to perfect foresight.

Two checks say when a linear impulse response has changed the answer. The z-transform compares the nonlinear and log-linear responses on the unit circle, in magnitude and in phase, including a shock announced before it arrives. That shows which frequencies of the news the linear response misses. The second check adds up pieces of a nonlinear response: the anticipatory part and the on-impact part of a one-period shock are superposed into a multi-period path and compared with the true path for the same sequence. A large gap on the unit circle, or a failure of that superposition, means the linear step has changed the economics of the announcement.